ANSWERS

1 **(a)** transfer / propagation of energy M1
as a result of oscillations / vibrations .... A1 [2]

(b) **(i)** displacement / velocity / acceleration (of particles in the wave) .... B1 [1]

(ii) displacement etc. is normal to direction of energy transfer /
travel of wave / propagation of wave ……(*not 'wave motion'*) .... B1 [1]

(iii) displacement etc. along / same direction of energy transfer /
travel of wave / propagation of wave ……(*not 'wave motion'*) .... B1 [1]

(c) diffraction: suitable object, means of observation M1
either laser *or* lamp and aperture
or distant source M1
light region where darkness expected A1
interference: suitable object, means of observation and illumination B1
light and dark fringes observed B1
appropriate reference to a dimension for diffraction or
for interference .... B1 [6]

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2 **(a)** **(i)** frequency f .... B1 [1]

(ii) amplitude A .... B1 [1]

(b) π rad or 180° ………(unit necessary) .... B1 [1]

(c) **(i)** speed = $f \times L$ .... B1 [1]

(ii) wave is reflected at end / at P B1
either incident and reflected waves interfere
or two waves travelling in opposite directions interfere M1
speed is the speed of incident or reflected wave / one of these waves .... A1 [3]

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3 **(a)** e.g. no energy transfer
amplitude varies along its length/nodes <u>and</u> antinodes
neighbouring points (in inter-nodal loop) vibrate in phase, etc.
(*any two, 1 mark each to max 2* ……………………………………………………….B2 [2]

(b) **(i)** $\lambda = (330 \times 10^2)/550$ …………………………………………………………. M1
λ = 60 cm ………………………………………………………………… A0 [1]

(ii) node labelled at piston ………………………………………………… B1
antinode labelled at open end of tube ……………………………………… B1
additional node and antinode in correct positions along tube …………… B1 [3]

(c) at lowest frequency, length = $\lambda/4$ ………………………………………………... C1
λ = 1.8 m
frequency = 330/1.8 ………………………………………………………………. C1
= 180 Hz ……………………………………………………………………….... A1 [3]

4	**(a)**	**(i)**	distance (of point on wave) from rest / equilibrium position	B1	[1]
		(ii)	distance moved by wave energy / wavefront during one cycle of the source **or** minimum distance between two points with the same phase **or** between adjacent crests or troughs	B1	[1]
	(b)	**(i)**	$T = 0.60\,\text{s}$	B1	[1]
		(ii)	$\lambda = 4.0\,\text{cm}$	B1	[1]
		(iii)	*either* $v = \lambda / T$ *or* $v = f\lambda$ <u>and</u> $f = 1/T$	C1	
			$v = 6.7\,\text{cm}\,\text{s}^{-1}$	A1	[2]
	(c)	**(i)**	amplitude is decreasing	M1	
			so, it is losing power	A1	[2]
		(ii)	*intensity* ~ (*amplitude*)^2	C1	
			ratio = $2.0^2 / 1.1^2$	C1	
			= 3.3	A1	[3]

5	**(a)**		two waves travelling (along the same line) in opposite directions overlap/meet	M1	
			same frequency / wavelength	A1	
			resultant displacement is the sum of displacements of each wave / produces nodes and antinodes	B1	[3]
	(b)		apparatus: source of sound + detector + reflection system	B1	
			adjustment to apparatus to set up standing waves – how recognised	B1	
			measurements made to obtain wavelength	B1	[3]
	(c)	**(i)**	at least two nodes and two antinodes	A1	[1]
		(ii)	node to node = $\lambda / 2 = 34\,\text{cm}$ (allow 33 to 35 cm)	C1	
			$c = f\lambda$	C1	
			$f = 340 / 0.68 = 500$ (490 to 520) Hz	A1	[3]

6	**(a)**			waves (travels along tube) reflect at <u>closed end / end of tube</u>	B1	
				incident and reflected waves or these two waves are in <u>opposite directions</u>	M1	
				interfere or stationary wave formed if tube length equivalent to $\lambda / 4$, $3\lambda / 4$, etc.	A1	[3]
	(b)	**(i)**	**1.**	no motion (as node) / zero amplitude	B1	[1]
			2.	vibration backwards and forwards / maximum amplitude along length	B1	[1]
		(ii)		$\lambda = 330 / 880$ (= 0.375 m)	C1	
				$L = 3\lambda / 4$	C1	
				$L = 3 / 4 \times (0.375) = 0.28$ (0.281) m	A1	[3]

7	**(a)**	travel through a vacuum / free space	B1	[1]
	(b) (i)	B : name: **microwaves** wavelength: 10^{-4} to 10^{-1} m	B1	
		C : name: **ultra-violet / UV** wavelength: 10^{-7} to 10^{-9} m	B1	
		F : name: **X –rays** wavelength: 10^{-9} to 10^{-12} m	B1	[3]
	(ii)	$f = \dfrac{3 \times 10^{8}}{500 \times 10^{-9}}$	C1	
		$f = 6(.0) \times 10^{14}$ Hz	A1	[2]
	(c)	vibrations are in one direction	M1	
		perpendicular to direction of propagation / energy transfer or good sketch showing this	A1	[2]
8	**(a) (i)**	$v = f\lambda$	C1	
		$\lambda = 40 / 50 = 0.8(0)$ m	A1	[2]
	(ii)	waves (travel along string and) reflect at Q / wall / fixed end	B1	
		incident and reflected waves interfere / superpose	B1	[2]
	(b) (i)	nodes labelled at P, Q and the two points at zero displacement	B1	
		antinodes labelled at the three points of maximum displacement	B1	[2]
	(ii)	(1.5λ for PQ hence PQ = 0.8 × 1.5) = 1.2 m	A1	[1]
	(iii)	$T = 1 / f = 1/50 = 20$ ms	C1	
		5 ms is ¼ of cycle	A1	
		horizontal line through PQ drawn on Fig. 5.2	B1	[3]
9	**(a) (i)**	displacement is the distance the rope / particles are (above or below) from the equilibrium / mean / rest / undisturbed position (not 'distance moved')	B1	[1]
	(ii) 1.	amplitude (= 80 / 4) = 20 mm	B1	[1]
	2.	$v = f\lambda$ or $v = \lambda / T$	C1	
		$f = 1 / T = 1 / 0.2$ (5 Hz)	C1	
		$v = 5 \times 1.5 = 7.5$ m s^{-1}	A1	[3]
	(b)	point A of rope shown at equilibrium position	B1	
		same wavelength, shape, peaks / wave moved ¼λ to right	B1	[2]
	(c) (i)	progressive as energy OR peaks OR troughs is/are transferred/moved /propagated (by the waves)	B1	[1]
	(ii)	transverse as particles/rope movement is perpendicular to direction of travel /propagation of the energy/wave velocity	B1	[1]

10 (a) (i)	**1.** wavelength: minimum distance between two points moving in phase OR distance between neighbouring or consecutive peaks or troughs OR wavelength is the distance moved by a wavefront in time *T* or one oscillation/cycle or period (of source)	B1	[1]
	2. frequency: number of wavefronts / (unit) time OR number of oscillations per unit time or oscillations/time	B1	[1]
(ii)	speed = <u>distance</u> / time = <u>wavelength / time period</u> $= \lambda / T = \lambda f$	M1 A0	[1]
(b) (i)	amplitude = 4.0 mm (allow 1 s.f.)	A1	[1]
(ii)	wavelength = 18 / 3.75 (= 4.8)	C1	
	speed = $2.5 \times 4.8 \times 10^{-2} = 12 \times 10^{-2}\,\mathrm{m\,s^{-1}}$ unit consistent with numerical answer, e.g. in $\mathrm{cm\,s^{-1}}$ if cm used for λ and unit changed on answer line	A1	[2]
	[if 18 cm = 3.5λ used giving speed 13 (12.9) $\mathrm{cm\,s^{-1}}$ allow max. 1].		
(iii)	180° or π rad	A1	[1]
(c)	light and screen and correct positions above and below ripple tank strobe or video camera	B1 B1	[2]

11 (a) (i)	displacement is the distance from the equilibrium position / undisturbed position / midpoint / rest position amplitude is the maximum displacement	B1 B1	[2]
(ii)	frequency is the number of wavefronts / crests passing a point per unit time / number of oscillations per unit time time period is the time between adjacent wavefronts *or* time for one oscillation	B1 B1	[2]
(b) (i)	**1.** amplitude = 1.5 mm	A1	[1]
	2. wavelength = 25 / 6	C1	
	= 4.2 cm or 4.2×10^{-2} m	A1	[2]
(ii)	$v = \lambda / T$ *or* $v = f\lambda$ <u>*and*</u> $T = 1/f$ $T = 4.2 / 7.5 = 0.56$ s	C1 A1	[2]
(c) (i)	progressive wavefront / crests moving / energy is transferred by the waves	M0 A1	[1]
(ii)	transverse the vibration is perpendicular to the direction of energy transfer / wave velocity *or* travel <u>of the wave / wavefronts</u>	M0 A1	[1]

12	**(a)**	**(i)**	progressive wave transfers energy, stationary wave no transfer of energy / keeps energy within wave	B1	[1]
		(ii)	(progressive) wave / wave from loudspeaker reflects at end of tube	B1	
			reflected wave overlaps (another) progressive wave	B1	
			same frequency and speed hence stationary wave formed	B1	[3]
		(iii)	(side to side) along length of tube / along axis of tube	B1	[1]
	(b)		all three nodes clearly marked with N / clearly labelled at cross-over points	B1	[1]
	(c)		phase difference = 0	A1	[1]
	(d)	**(i)**	$v = f\lambda$	C1	
			$\lambda = 330/440 = 0.75\,\text{m}$	A1	[2]
		(ii)	$L = 5/4\ \lambda$	C1	
			$= 5/4 \times 0.75 = 0.94\,\text{m}$	A1	[2]

13	**(a)**		two waves (of the same kind) travelling in opposite directions overlap	B1	
			waves have same frequency / wavelength and speed	B1	[2]
	(b)	**(i)**	$T = 0.8\,(\text{ms})$	C1	
			$f = 1 / (0.8 \times 10^{-3}) = 1250\,(\text{Hz})$	A1	[2]
		(ii)	microphone is moved from plate to loudspeaker or vice versa	B1	
			wavelength is the twice the distance between adjacent maxima or minima (seen on c.r.o.)	B1	[2]
		(iii)	$v = f\lambda$	C1	
			$= 1250 \times 0.26$		
			$= 330\ (325)\,\text{m}\,\text{s}^{-1}$	A1	[2]

14	**(a)**		stress = Young modulus $\times$ strain		
			$= 1.8 \times 10^{11} \times 8.2 \times 10^{-4}$ *or* 1.476×10^{8}	C1	
			$= 0.15\ (0.148)$ GPa	A1	[2]
	(b)	**(i)**	wavelength $= 3 \times 10^{8} / 12 \times 10^{12}$	C1	
			$= 25\,\mu\text{m}$	A1	[2]
		(ii)	infra-red / IR	B1	[1]

15	**(a)**	**(i)**	progressive: energy is moved / transferred / propagated from one place to another (without the bulk movement of the medium)	B1	
			transverse: (particles) oscillate / vibrate at right angles to the direction of travel of the energy / wavefront	B1	[2]
		(ii)	number of oscillations per unit time / number of wavefronts passing a point per unit time	B1	[1]
	(b)	**(i)**	P and T	B1	[1]
		(ii)	P and S <u>or</u> Q and T	B1	[1]
	(c)		$\lambda = 1.2 \times 10^{-2}$ (m)	C1	
			$v = f\lambda \quad = 15 \times 1.2 \times 10^{-2}$	C1	
			$= 0.18\,\mathrm{m\,s^{-1}}$	A1	[3]
	(d)		ratio $= (1.4)^2 / (2.1)^2$	C1	
			$= 0.44$	A1	[2]
16	**(a)**		progressive waves transfer/propagate energy **and** stationary waves do not	B1	
			<u>amplitude</u> constant for progressive wave **and** varies (from max/antinode to min/zero/node) for stationary wave	B1	
			adjacent particles in phase for stationary wave **and** out of phase for progressive wave	(B1)	[2]
	(b)	**(i)**	wave / microwave <u>from source/S</u> reflects <u>at reflector/R</u>	B1	
			reflected and (further) incident waves overlap/meet/superpose	B1	
			waves have same <u>frequency/wavelength/period</u> **and** <u>speed</u> (so stationary waves formed)	B1	[3]
		(ii)	detector/D is moved between reflector/R and source/S (or v.v.)	B1	
			maximum, minimum/zero, (maximum… etc.) observed on meter/deflections/readings/measurements/recordings	B1	[2]
		(iii)	<u>determine/measure</u> the distance between adjacent minima/nodes or maxima/antinodes **or** across <u>specific number</u> of nodes/antinodes	B1	
			wavelength is twice distance between <u>adjacent</u> nodes/minima or maxima/antinodes (or other correct method of calculation of wavelength from measurement)	B1	[2]
	(c)		$v = f\lambda$	C1	
			$f = 3.0 \times 10^8 / (2.8 \times 10^{-2})$ [= 1.07×10^{10} Hz]	C1	
			11 (10.7) GHz	A1	[3]
17		**(i)**	intensity $\propto$ (amplitude)^2	C1	
			ratio $= (0.60^2 / 0.90^2) = 0.44$	A1	
		(ii)	phase difference = 90°	A1	
18	**(a)**		$T = 4$ (ms) or 4×10^{-3} (s)	C1	
			$f = 1/T = 1/0.004$		
			$= 250\,\mathrm{Hz}$	A1	[2]
	(b)		intensity $\propto$ (amplitude)^2 <u>and</u> amplitude = 2.8 (2.83) (cm)	B1	
			curve with same period and with amplitude 2.8 cm	B1	
			curve shifted 1.0 ms to left or to right of wave X	B1	[3]

19 (a) longitudinal: vibrations/oscillations (of the particles/wave) are parallel to the direction **or** in the same direction (of the propagation of energy)	B1	
transverse: vibrations/oscillations (of the particles/wave) are perpendicular to the direction (of the propagation of energy)	B1	[2]
(b) LHS: intensity = power / area units: $kg\,m\,s^{-2} \times m \times s^{-1} \times m^{-2}$ or $kg\,m^2 s^{-3} \times m^{-2}$	B1	
RHS: units: $m\,s^{-1} \times kg\,m^{-3} \times s^{-2} \times m^2$	M1	
LHS and RHS both $kg\,s^{-3}$	A1	[3]
(c) (i) change/difference in the <u>observed</u>/<u>apparent</u> frequency when the source is moving (relative to the observer)	B1	[1]
(ii) wavelength increases/frequency decreases/red shift	B1	[1]
(d) observed frequency = $vf_S / (v - v_S)$	C1	
$550 = (340 \times 510) / (340 - v_S)$	C1	
$v_S = 25\ (24.7)\ m\,s^{-1}$	A1	[3]
20 (a) (i) alter distance from vibrator to pulley alter frequency of generator (change tension in string by) changing value of the masses *any two*	B2	[2]
(ii) points on string have <u>amplitudes</u> varying from maximum to zero/minimum	B1	[1]
(b) (i) 60° or $\pi / 3$ rad	A1	[1]
(ii) ratio = $[3.4 / 2.2]^2$	C1	
= 2.4 (2.39)	A1	[2]
21 (a) the number of oscillations per unit time	M1	
of the source/of a point on the wave/of a particle (in the medium)	A1	[2]
or the number of wavelengths/wavefronts per unit time	(M1)	
passing a (fixed) point	(A1)	
(b) *T* or period = 2.5×250 (μs) (= 625 μs)	M1	
frequency = $1 / (6.25 \times 10^{-4})$ or $1 / (2.5 \times 250 \times 10^{-6})$ = 1600 Hz	A1	[2]
(c) (i) for maximum frequency: $f_o = f_s v / (v - v_s)$		
$1640 = (1600 \times 330) / (330 - v_s)$	C1	
$v_s = 8(.0)\ m\,s^{-1}\ (8.049\ m\,s^{-1})$	A1	[2]
(ii) loudspeaker moving towards observer causes rise in/high<u>er</u> frequency	B1	
loudspeaker moving away from observer causes fall in/low<u>er</u> frequency	B1	[2]
or		
<u>repeated</u> rise and fall/higher and then lower frequency	(M1)	
caused by loudspeaker moving towards and away from observer	(A1)	

22	**(a)** (two) waves travelling (at same speed) in opposite directions overlap	**B1**
	waves (are same type and) have same frequency/wavelength	**B1**
	(b) (i) $\lambda = 12 / 250$ (= 0.048 m)	**C1**
	distance $= 1.5 \times 0.048$	**A1**
	$= 0.072$ m	
	(ii) $T = 1 / 250$	**C1**
	= 0.004 (s) or 4 (ms)	
	1. curve drawn is mirror image of that in Fig. 4.2 and labelled P	**A1**
	2. horizontal line drawn between A and B and labelled Q	**A1**
23	**(a)** observed frequency is different to source frequency when source moves relative to observer	**B1**
	(b) $360 = (400 \times 340) / (340 \pm v)$	**C1**
	$v = 38\ (37.8)\ \text{m s}^{-1}$	**A1**
	away (from the observer)	**B1**
24	**(a)** frequency is the number of vibrations/oscillations per unit time **or** the number of wavefronts passing a point per unit time	
	(b) vibrations/oscillation of the air particles are parallel to the direction of it (the direction of travel of the sound wave)	**B1**
	(c) (i) T = 2(.0) (ms) $f = 500$ Hz	**A1**
	(ii) 1. amplitude increases (time) period decreases	**B3**
	2. amplitude decreases (time) period increases	*any 3 points*
25	**(a) (i)** in a stationary wave energy is not transferred **or** in a progressive wave energy is transferred	**B1**
	(ii) in a stationary wave (adjacent) particles are in phase **or** in a progressive wave (adjacent) particles are out of phase/have a phase difference/not in phase	**B1**
	(b) (i) (position where) maximum amplitude	**B1**
	(ii) distance = 0.10 m	**B1**
	(iii) 1. $\lambda = 0.60/1.5$	**A1**
	$= 0.40$ m	
	2. $v = f\lambda$	**C1**
	$f = 340/0.40$	**A1**
	$= 850$ Hz	
	(iv) $\lambda = 2 \times 0.60$ **or** $\lambda = 3 \times 0.40$ **or** $f = 850/3$	**C1**
	$f = 280\ (283)$ Hz	**A1**

26 **(a)**	(two) waves travelling (at same speed) in opposite directions overlap	**B1**
	waves (are same type and) have same frequency/wavelength	**B1**
(b) (i)	5	**A1**
(ii)	$T = 1/40$ (= 2.5×10^{-2})	**C1**
	time taken = $2.5 \times 10^{-2}/2$	**A1**
	= 1.3×10^{-2} s (1.25×10^{-2} s)	
(iii)	180°	**A1**
(iv)	$v = f\lambda$	**C1**
	$\lambda = 2.0/2.5$ (= 0.80 m)	**A1**
	$v = 0.80 \times 40$	
	= 32 m s^{-1}	

27 **(a)**	displacement of particles/vibration(s)/oscillation(s) is parallel to/along the direction of energy/propagation	**B1**
(b)	period = 1 / 800 (= 1.25×10^{-3}s)	**C1**
	time-base setting = 1.25×10^{-3} / 2.5	**C1**
	= 5.0×10^{-4} s cm^{-1}	**A1**
(c) (i)	$I \propto A^2$	**C1**
	$(I_X / I_Y =) [r_Y / r_X]^2 = [A_X / A_Y]^2$	**C1**
	ratio A_Y / A_X = 120 / 30	**A1**
	= 4.0	
(ii)	**1.** $v = f\lambda$	**C1**
	minimum λ = 330 / (800 + 16) = 0.40 m	**A1**
	2. $f_o / f_s = v / (v - v_s)$	**C1**
	$816 / 800 = 330 / (330 - v_s)$	
	v_s = 6.5 m s^{-1}	**A1**

28 (a) (i) time for one oscillation/one vibration/one cycle **B1**
or time between adjacent wavefronts/points in phase
or shortest time between two wavefronts/points in phase

(ii) distance moved by wavefront/energy during one cycle/oscillation/period (of source) **B1**
or minimum distance between two wavefronts
or distance between two adjacent wavefronts
or minimum distance between two points having the same displacement and moving in the same direction

(b) (i) $v = \lambda / T$ **or** $v = f\lambda$ <u>and</u> $f = 1/T$ **C1**
$\lambda = 20 \times 0.60$ **A1**
$= 12$ cm

(ii) phase difference $= 360° \times (0.20/0.60)$ **or** $360° \times (0.40/0.60)$ **A1**
$= 120°$ **or** $240°$

(iii) $I \propto A^2$ **C1**
$I_Q/I_P = A_Q^{\,2}/A_P^{\,2}$ **A1**
$= 2.0^2/3.0^2$
$= 0.44$

(iv) displacement $= 1.00 - 3.00$ **A1**
$= -2.00$ mm

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29 (a) (i) distance moved by wavefront/energy during one cycle/oscillation/period (of source) **B1**
or <u>minimum</u> distance between two wavefronts
or distance between two <u>adjacent</u> wavefronts

(ii) (position where) maximum amplitude **B1**

(b) (i) $\lambda = 4 \times 0.045$ **C1**
(= 0.18 (m) or 18 (cm))
$v = f\lambda$ **C1**
$f = 340/0.18$ **A1**
$= 1900$ Hz

(ii) distance $= \lambda/2$ **C1**
(= 0.09 (m) or 9 (cm))
time = 0.09 / 0.0075 **A1**
= 12 s

or $t = 4.5/0.75$ **and** $t = 13.5/0.75$ **(C1)**
time = 18 – 6 **(A1)**
= 12 s

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30 (a)	vibration(s)/oscillation(s) (of particles) parallel to direction of propagation of energy	**B1**
(b)(i)	phase difference = 180°	**A1**
(ii)	$v = f\lambda$	**C1**
	$\lambda/2 = 25$ (cm) or 0.25 (m)	**C1**
	$f = 330/0.50$	**A1**
	= 660 Hz	
(iii)	(readings from graph =) 2.6 and 4.0	**C1**
	ratio = $(2.6/4.0)^{1/2}$	**A1**
	= 0.81	

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31 (a)	$\rho = m / V$	**C1**
	$V = \pi \times (0.16 / 2)^2 \times 7.6 \times 3.0$ (= 0.458 m^3)	**C1**
	$m = \pi \times (0.16 / 2)^2 \times 7.6 \times 3.0 \times 1.2 = 0.55$ kg	**A1**
(b)(i)	$\Delta p = 0.55 \times 7.6$	**A1**
	= 4.2 N s	
(ii)	$F = 4.2 / 3.0$ **or** $0.55 \times 7.6 / 3.0$	**A1**
	= 1.4 N	
(c)(i)	$F = 1.4$ N	**A1**
(ii)	Newton's third law (of motion)	**B1**
(d)	$2 \times 1.4 = m \times 9.81$	**A1**
	$m = 0.29$ kg	
(e)	the density of air is less at high altitude	**B1**
(f)	$f_o = f_s v / (v - v_s)$	**C1**
	$= 3000 \times 340 / (340 - 22)$	
	= 3200 Hz	**A1**

32	(a)	distance moved by wavefront/energy during one cycle/oscillation/period (of source) **or** <u>minimum</u> distance between two wavefronts **or** distance between two <u>adjacent</u> wavefronts	**B1**
	(b)	(T =) 2.0×2.5 (= 5.0 ms) **or** $2.0 \times 2.5 \times 10^{-3}$ (= 5.0×10^{-3} s) $f = 1 / (5.0 \times 10^{-3})$ = 200 Hz	**C1** **A1**
	(c)(i)	(path difference =) $8.0 + (20.8^2 - 8.0^2)^{0.5} - 20.8 = 6.4$ (m)	**A1**
	(ii)	• <u>path difference</u> = 4λ • waves (meet at C) in phase • constructive interference (of waves) *any two points, one mark each*	**B2**
	(iii)	$v = 200 \times 1.6$ = 320 (m s^{-1}) $\Delta t = 6.4 / 320$ **or** $27.2 / 320 - 20.8 / 320$ = 0.020 s	**C1** **A1**
	(iv)	$3\lambda = 6.4$ $\lambda = 2.1$ m	**A1**

33	(a)(i)	frequency **or** period	**B1**
	(ii)	amplitude	**B1**
	(b)	constant phase difference so coherent	**B1**
	(c)	120°	**B1**
	(d)	resultant displacement = 4.0 µm – 1.0 µm = 3.0 µm	**B1**
	(e)	$I \propto A^2$ intensity of Z = $(2^2 / 4^2)\ I$ = $0.25\ I$	**C1** **A1**
	(f)	$v = \lambda / T$ **or** $v = f\lambda$ **and** $f = 1 / T$ $330 = \lambda / 3.0 \times 10^{-3}$ $\lambda = 0.99$ m	**C1** **C1** **A1**

34.

5(a)	$v = \lambda / T$ **or** $v = f\lambda$ **and** $f = 1 / T$	**C1**
	$v = 8.0 \times 10^{-2} / 0.40$ $= 0.20\text{ m s}^{-1}$	**A1**
5(b)	$I \propto A^2$	**C1**
	ratio $= 2^2 / 4^2$	**C1**
	$= 0.25$	**A1**

35.

5(a)	$f_o = f_s v / (v + v_s)$ $f_o = 951 \times 330 / (330 + 12)$	**C1**
	$= 918\text{ Hz}$	**A1**
5(b)	$t = d / 12$ $= (\pi \times 2.4) / 12$	**C1**
	$= 0.63\text{ s}$	**A1**

36.

4(a)	oscillations (of particles) are parallel to (the direction of) energy transfer	**B1**
4(b)(i)	(frequency varies as) vehicle moves relative to (stationary) observer	**C1**
	(vehicle) moving towards (observer) gives higher (observed) frequency (than 1.2 kHz) <u>and</u> (vehicle) moving away (from observer) gives lower (observed) frequency (than 1.2 kHz)	**A1**
4(b)(ii)	Doppler effect	**B1**
4(b)(iii)	position of vehicle labelled ‘X’ at top (12 o’clock) position on track	**B1**
4(b)(iv)	position of vehicle labelled ‘Y’ at right-hand edge (3 o’clock) position on track	**B1**
4(c)	maximum frequency = 1.40 (kHz) **or** 1.40×10^3 (Hz)	**C1**
	$1.40 = (1.2 \times 320) / (320 - v)$	**C1**
	$v = 46\text{ m s}^{-1}$	**A1**
	or	
	minimum frequency = 1.05 (kHz) **or** 1.05×10^3 (Hz)	**(C1)**
	$1.05 = (1.2 \times 320) / (320 + v)$	**(C1)**
	$v = 46\text{ m s}^{-1}$	**(A1)**

37.

4(a)(i)	decrease(s)	**B1**
4(a)(ii)	increase(s)	**B1**
4(b)	$f_o = f_s v / (v + v_s)$ $= 925 \times 338 / (338 + 12)$	**C1**
	$= 893$ Hz	**A1**
4(c)	distance $= (½ \times 2 \times 12) + (2 \times 12)$ $(= 36$ m$)$	**C1**
	time taken $= 36 / 338$ $= 0.11$ s	**A1**

38.

5(a)(i)	(two) waves travelling (at same speed) in opposite directions overlap	**B1**
	waves (of the same type) have same frequency/wavelength	**B1**
5(a)(ii)	phase difference = 0	**A1**
5(b)(i)	$f_O = f_S v / v – v_S$ $543 = f \times 334 / (334 – 13)$	**C1**
	$f = 522$ Hz	**A1**
5(b)(ii)	(the speed is) decreasing	**B1**
5(c)(i)	$I \propto A^2$	**B1**
	$I_T / I_0 = \cos^2 20°$ or $A_T / A_0 = \cos 20°$	**C1**
	ratio = 0.94	**A1**
5(c)(ii)	angle = 140°	**A1**

39.

4(a)(i)	oscillations are in a single direction, which is perpendicular to the direction of propagation (of the wave) **or** oscillations are in a single plane, which contains the direction of propagation (of the wave)	**B1**
4(a)(ii)	light waves are transverse **and** sound waves are longitudinal	**B1**
4(b)	$I = I_0 \cos^2\theta$	**C1**
	$\cos^2 \theta = 1 / 4$ so $\cos \theta = 1 / 2$ $\theta = 60°$ or $120°$ or $240°$ or $300°$	**C1**
	angle of rotation $= (120° – 90°)$ or $(240° – 90°)$ or $(300° – 90°)$ $= 30°$ or $150°$ or $210°$ or $330°$	**A1**
4(c)(i)	the waves have different amplitudes	**B1**
	cannot have resultant displacement that is always zero **or** cannot have (complete) destructive interference (at nodes) **or** (at nodes resultant) amplitude is the difference of the amplitudes	**B1**
4(c)(ii)	$I \propto A^2$	**C1**
	$A^2 / A_0^2 = (I_0 / 4) / I_0$ $A = 0.5 A_0$	**C1**
	maximum amplitude $= A_0 + 0.5 A_0$ $= 1.5 A_0$	**A1**

40.

5(a)	(they travel in) opposite directions	**B1**
5(b)(i)	straight line from A to B, labelled P	**B1**
	line that is 'mirror image' of given line, labelled Q	**B1**
5(b)(ii)	$\lambda = 0.80 / 2$ $(= 0.40\ \text{m})$	**C1**
	$v = \lambda / T$ **or** $v = f\lambda$ **and** $f = 1 / T$	**C1**
	$v = 0.40 / 0.016$ **or** 62.5×0.40 $= 25\ \text{m s}^{-1}$	**A1**
5(c)(i)	$I_1 / I_0 = \cos^2 30°$	**C1**
	$= 0.75$	**A1**
5(c)(ii)	$I_2 / I_1 = \cos^2 60°$ $I_2 / I_0 = \cos^2 30° \times \cos^2 60°$ **or** $0.75 \times \cos^2 60°$	**C1**
	$= 0.19$	**A1**

41.

4(a)(i)	circles drawn around any two particles with seven (uncircled) particles in between	**A1**
4(a)(ii)	curve has an initial negative displacement and initial amplitude same as original curve	**B1**
	curve has same amplitude as original curve throughout	**B1**
	curve has same wavelength as original curve throughout, with constant (non-zero) phase difference	**B1**
4(a)(iii)	(to the) right / rightwards	**A1**
4(b)	$\lambda = 2 \times 0.19$ $= 0.38\ \text{m}$	**C1**
	$v = f\lambda$	**C1**
	$v = (16 \times 10^3) \times 0.38$ $v = 6100\ \text{m s}^{-1}$	**A1**
4(c)	$I \propto A^2$	**C1**
	$I = (1 / 2)^2\ I_0$ intensity $= 0.25\ I_0$	**A1**
4(d)(i)	same frequency / wavelength / period	**B1**
4(d)(ii)	• a stationary wave has nodes/antinodes (and a progressive wave does not) • a stationary wave does not transfer/propagate energy (and a progressive wave does transfer/propagate energy) • different points on a stationary wave have different amplitudes (and all points on a progressive wave have the same/constant amplitude) • stationary wave has adjacent particles that are in phase (and adjacent particles on progressive wave are out of phase) *Any two points, 1 mark each. Allow the reverse statement for each marking point.*	**B2**

42.

5(a)(i)	(they are) perpendicular	**B1**
5(a)(ii)	(they are) parallel	**B1**
5(b)(i)	$\lambda = v / f$	**C1**
	$= 340 / 1700$ $= 0.20$ m	**A1**
5(b)(ii)	$L = \frac{3}{4} \times \lambda = \frac{3}{4} \times 0.20$ $= 0.15$ m	**A1**
5(b)(iii)	$\lambda = 4 \times 0.15$ **or** 0.20×3 $= 0.60$ m	**A1**
5(c)(i)	(I =) $8.5 \times \cos^2 35° = 5.7$ (W m^{-2})	**A1**
5(c)(ii)	$5.2 = 5.7 \cos^2 \theta$ ($\theta = 17°$)	**C1**
	$\alpha = 35° + 17°$ $= 52°$	**A1**

43.

5(a)	sketch: approximately horizontal line above horizontal dashed line from $t = 0$ to $t = t_1$ **and** approximately horizontal line below horizontal dashed line from $t = t_1$ to $t = t_2$	**A1**
5(b)	$I = P / A$	**C1**
	$A = \pi \times 0.028^2$ **or** $\pi \times 2.8^2$ ($= 2.46 \times 10^{-3}$ **or** 24.6)	**C1**
	$P = 4.7 \times 10^{-3} \times 2.46 \times 10^{-3}$ $= 1.2 \times 10^{-5}$ W	**A1**

44.

6(a)	$291 = 251 \times 340 / (340 - v_{(s)})$	**C1**
	$v_s = 47$ m s^{-1}	**A1**
6(b)(i)	$I \propto A^2$	**B1**
6(b)(ii)	sketch: line starts at (1.0, 1.0)	**B1**
	approximately straight line drawn with negative gradient and line ends at (4.0, 0.25)	**B1**